26 B By the principle of conservation of energy,
Gain in KE of particle = Loss in Electric Potential Energy
$\frac{1}{2} mv^2 - 0 = qV$

$$v = \sqrt{\frac{2qV}{m}} \text{ ---------- (1)}$$

As magnetic force provides centripetal force,

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq} \text{ ---------- (2)}$$

Substitute (1) into (2):

$$r = \frac{m}{Bq}\sqrt{\frac{2qV}{m}} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$$

Since B and V are the same for both,

$$r \propto \sqrt{\frac{m}{q}}$$

$$\frac{r_Y}{r_X} = \sqrt{\frac{m_Y q_X}{m_X q_Y}} = \sqrt{\frac{m_Y}{m_X} \times \frac{q_X}{q_Y}} = \sqrt{2 \times \frac{1}{4}}$$

Therefore, $r_Y = \frac{r_X}{\sqrt{2}}$

27 B *F = Magnetic force on YZ + Magnetic force on XY*